J1A
nuMezzanine_4x34
TCLK 23 TCLK
T00 32 T0_UNUSED
T01 31 T0_UNUSED
T02 30 T0_UNUSED
T03 29 T0_UNUSED
T04 28 T0_UNUSED
T05 27 T0_UNUSED
T06 26 T06
T07 25 T07
T10 21 T1_UNUSED
T11 20 T1_UNUSED
T12 19 T12
T13 18 T13
T14 17 T14
T15 16 T15
T16 15 T16
T17 14 T1_UNUSED
T20 9 T20
T21 8 T21
T22 7 T22
T23 6 T23
T24 5 T24
T25 4 T25
T26 3 T26
T27 2 T27
TX0 13
TX1 12
TX2 11
TX3 10

J1C
nuMezzanine_4x34
BCLK 91
B00 100 B0_UNUSED
B01 99 B0_UNUSED
B02 98 B02
B03 97 B03
B04 96 B04
B05 95 B05
B06 94 B06
B07 93 B07
B10 89 B10
B11 88 B11
B12 87 B12
B13 86 B13
B14 85 B14
B15 84 B15
B16 83 B16
B17 82 B17
B20 77 B20
B21 76 B21
B22 75 B22
B23 74 B2_UNUSED
B24 73 B2_UNUSED
B25 72 B2_UNUSED
B26 71 B2_UNUSED
B27 70 B2_UNUSED
BX0 81
BX1 80
BX2 79
BX3 78

J1E
nuMezzanine_4x34
VCC_TL 1 VCC_TL
VCC_BL 103
VCC_BR 69
VCC_TR 35
3V3_SELECT 34
5V_SELECT 68 GND
GND 33
GND

J1B
nuMezzanine_4x34
LCLK 125
L00 134 L00
L01 133 L01
L02 132 L02
L03 131 L03
L04 130 L04
L05 129 L05
L06 128 L06
L07 127 L07
L10 123 L10
L11 122 L11
L12 121 L12
L13 120 L13
L14 119 L14
L15 118 L15
L16 117 L16
L17 116 L17
L20 111 L20
L21 110 L21
L22 109 L22
L23 108 L23
L24 107 L24
L25 106 L25
L26 105 L26
L27 104 L27
LX0 115
LX1 114
LX2 113
LX3 112

J1D
nuMezzanine_4x34
RCLK 57 RCLK
R00 66 R0_UNUSED
R01 65 R0_UNUSED
R02 64 R0_UNUSED
R03 63 R03
R04 62 R04
R05 61 R05
R06 60 R06
R07 59 R07
R10 55 R10
R11 54 R11
R12 53 R12
R13 52 R13
R14 51 R14
R15 50 R15
R16 49 R16
R17 48 R17
R20 43 R20
R21 42 R21
R22 41 R22
R23 40 R23
R24 39 R2_UNUSED
R25 38 R2_UNUSED
R26 37 R2_UNUSED
R27 36 R2_UNUSED
RX0 47
RX1 46
RX2 45
RX3 44

DATA BUS

U1A
uPD70337L-8-xxx
L00 13 D0
L02 14 D1
L01 15 D2
L04 16 D3
L03 17 D4
L06 18 D5
L05 19 D6
L07 20 D7
D8 21 L10
D9 22 L12
D10 23 L11
D11 24 L14
D12 25 L13
D13 26 L16
D14 27 L15
D15 28 L17

MULTIPLEXED ADDRESS / BUS CONTROL

U1B
uPD70337L-8-xxx
L20 29 A0
L21 30 A9/A1
L22 31 A10/A2
L23 32 A11/A3
L26 33 A12/A4
L24 34 A13/A5
L27 35 A14/A6
L25 36 A15/A7
B03 37 A16/A8
B02 38 A17/A9
B05 39 A19
B04 40 A18/UBE
IOSTB 1 T15
MREQ 2 T16
MSTB 84 T14
R/W 83 T13
REFRQ 82 T12

PORT 0

U1C
uPD70337L-8-xxx
T20 4 P00
T21 5 P01
T22 6 P02
T23 7 P03
T24 8 P04
T25 10 P05
T26 11 P06
T27 12 P07/CLKOUT

PORT 1

U1D
uPD70337L-8-xxx
B15 59 P10/NMI
B17 60 P11/INTP0
B14 61 P12/INTP1
R21 62 P13/INTP2/INTAK
I07 63 P14/INT/POLL
R22 64 P15/TOUT
R23 65 P16/SCK0
T06 66 P17/READY

PORT 2 / DMA / ARBITRATION

U1E
uPD70337L-8-xxx
B20 48 P20/DMARQ0
R03 51 P21/DMAAK0
R04 52 P22/TCO
B21 54 P23/DMARQ1
R05 55 P24/DMAAK1
R07 56 P25/TC1
R06 57 P26/HLDAK
B22 58 P27/HLDRQ

SERIAL / RESET / CLOCK / MODE

U1F
uPD70337L-8-xxx
B10 41 RxD0
R20 43 CTS0/RSCK
B13 45 RxD1
B12 46 CTS1
B16 53 N/S
RCLK 78 X1
ICLK 77 X2
B11 81 RESET
TxD0 44 B06
TxD1 47 B07

PORT T / COMPARATOR REFERENCE

U1G
uPD70337L-8-xxx
R10 67 PT0
R12 68 PT1
R11 69 PT2
R14 70 PT3
R13 71 PT4
R16 72 PT5
R15 73 PT6
R17 74 PT7
VCC_TL 76 VTH

POWER / NEC RESERVED-PIN BIAS

U1H
uPD70337L-8-xxx
VCC_TL 50 VDD
VCC_TL 80 VDD
IC_BIAS 3 IC
IC_BIAS 49 IC
IC_BIAS 75 IC
RESERVED_PD 9 RESERVED
GND 42 GND
GND 77 GND

C1
100nF
VCC_TL
GND
At U1 pin 80

C2
100nF
VCC_TL
GND
At U1 pin 50

C3
100nF
VCC_TL
GND
At U1 pin 76

PWR_FLAG
GND

R1
6.8k
RESERVED_PD
GND
NEC reserved pin 9: 5–10k to GND

R2
15k
VCC_TL
IC_BIAS
NEC IC pins 3,49,75: shared 10–20k to VDD

HOST REQUIREMENTS
5V_SELECT is directly grounded: 3V3_SELECT is floating.
Drive valid inputs before DUT power-on: RESET is active LOW.
X1 and X2 require complementary phases of the same clock.
GPIO ports switch in compatible groups: see bank-modes.md.
Reserved-pin resistors follow NEC 4G-2: no bank-input pulls.
Suffix xxx is the unknown factory opcode translation-table code.

B0_UNUSED
R3
47k
GND
B0: B00, B01

B2_UNUSED
R4
47k
GND
B2: B23, B24, B25, B26, B27

R0_UNUSED
R5
47k
GND
R0: R00, R01, R02

R2_UNUSED
R6
47k
GND
R2: R24, R25, R26, R27

T0_UNUSED
R7
47k
GND
T0: T00, T01, T02, T03, T04, T05

T1_UNUSED
R8
47k
GND
T1: T10, T11, T17

UNUSED PINS IN USED BANKS: ONE SHARED 47k PULL-DOWN PER BANK
Drive all tied unused bits LOW together whenever host -> DUT, including startup.
See unused-bank-pins.csv for connector pad numbers.

uPD70337L-8-xxx / PLCC84 socket / 5 V / DMA and bus arbitration

Sheet: /
File: v35s.kicad_sch

Title: NEC V35S DUT daughter board

Size: A2	Date: 2026-09-13	**Rev: 1.0**
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